

Type of Rock at Giant's Causeway and Classification of Rock

The type of rock that makes up the majority of Giant's Causeway would be of the igneous rock type, specifically the black basalt columns that are created, which there are over 40 000 of them (*UNESCO*, 2025). These black basalt formations are most common inland, where there have been great amounts of lava that have been deposited to the Earth's surface, leading to the formation of the columns (*Regatta Universal Exports*, 2022). The reason why the black basalt is characterized as an igneous rock is because of the aphanitic texture that is present, where there are small/no crystals, which this type of basalt exhibits (see Figure 1). Additionally, the reason why the basalt rock is characterized as an igneous rock is because of how the formation of the columns of basalt came to be, through a process called *columnar jointing* (see Figure 2). Columnar jointing involves the cooling of lava flows. This cooling causes contractions, which creates fractures. The hexagonal pattern that is shown at Giant's Causeway is due to three fractures that are at 120 degrees from each other, which forms the hexagonal shape (*National Park Service*, 2023). Columnar jointing only happens with igneous rocks, and the basalt formations at Giant's Causeway are hexagonal in nature, so that further supports the reasoning of why it would be classified as an igneous rock. The history of the formation Giant's Causeway also provides evidence that the rocks present are igneous rocks, as there was evidence of volcanic activity from Europe ripping away from North America and creating rifts in Earth's surface. The rifts allowed for lava and molten rock to escape from the Earth's interior, which then crystallized into the basalt that is shown at Giant's Causeway today (*National Trust*, 2025). The reason why it wouldn't be a sedimentary rock would be because of the lack of fragmented rocks and minerals with cement holding it. The lack of halite and gypsum also strongly indicates why the rock formations wouldn't be classified as sedimentary rocks. The reason why the rock wouldn't be metamorphic rock would be because of the lack of visible layers, and the texture isn't foliated in nature.

Figure 1

Giant's Causeway Black Basalt Columns, courtesy of National Trust



Figure 1- The black basalt columns at Giant's Causeway exhibit minimal visible crystals and no layering/cementation of minerals

Figure 2

Visual Description of columnar jointing, courtesy of National Park Service

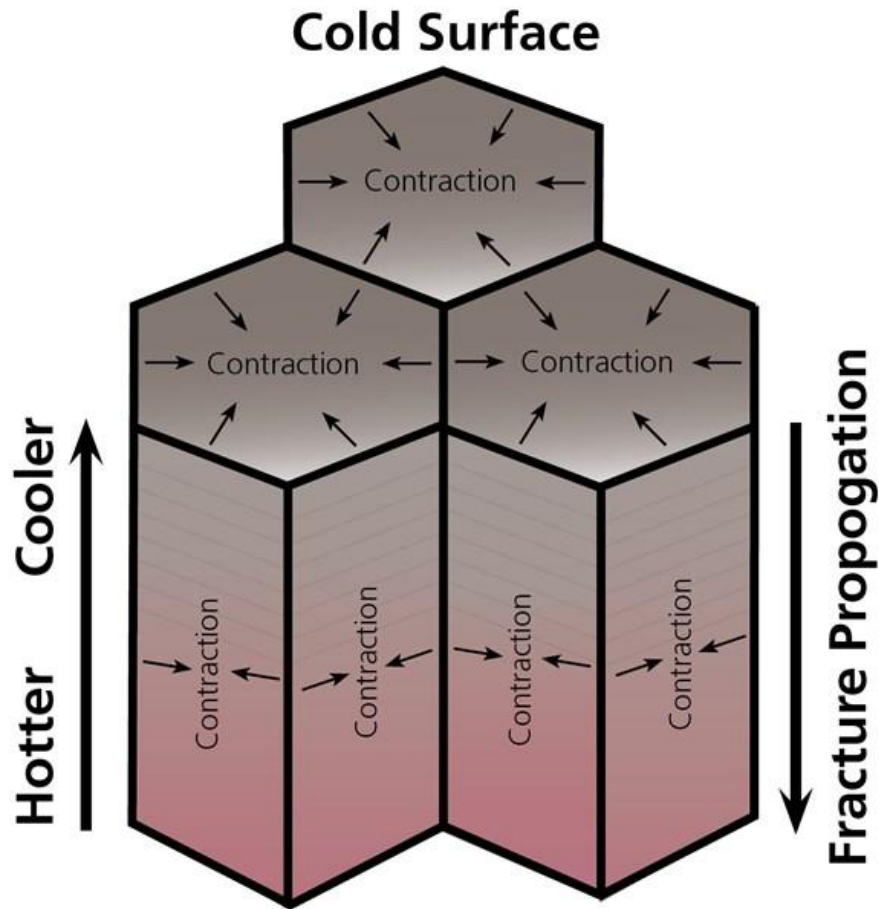


Figure 2- Description of the formation of the hexagonal basalt columns for Giant's Causeway

Adaptation of the Rock Cycle for the Formation of Basalt at Giant's Causeway

The adaptation of the rock cycle for the formation of the hexagonal basalt columns at Giant's Causeway would be that it started with magma rising from the surface when Europe separated from North America 60 million years ago (*National Trust*, 2025). When the magma cooled and crystallized, the basalt was created. As for the shape being hexagonal, that was through columnar jointing, where lava flowed cooled and created contractions. Through those contractions, fracturing happened, which is how the hexagonal formation happened (see Figure 2).

See Figure 3 below, which is the adaptation of the rock cycle based on the *thinkbluemarble* rock cycle diagram. I basically erased everything except for when the magma cools to form the igneous rock – which is what the basalt that is present at Giant's Causeway is a part of. The other parts of the rock cycle are extraneous and unnecessary due to the known history of Giant's Causeway with the lava flow and volcanic activity.

Figure 3

Modified rock cycle based on the history of Giant's Causeway. Adapted from thinkbluemarble website

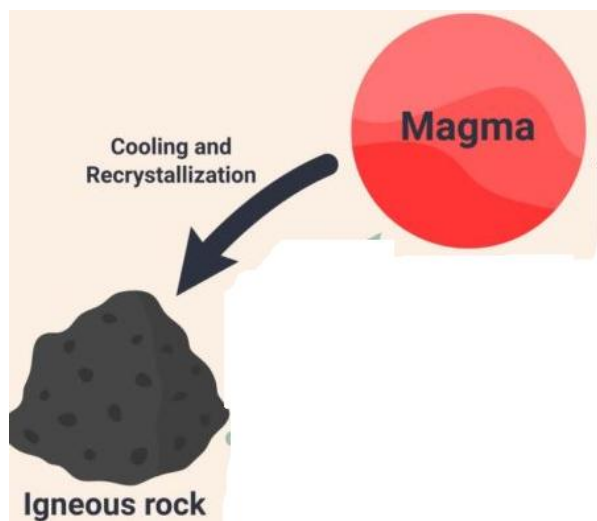


Figure 3 - Modified rock cycle. At Giant's Causeway, the magma that came from the rift created from the separation of Europe and North America cooled and crystallized into the basalt we see today. No other process from the rock cycle was involved in the formation.

Conclusions Based on Research Done on Giant's Causeway Thus Far

I've concluded that the formation of unique basalt columns from the lava flows at Giant's Causeway were caused through a special process with columnar jointing. Before researching columnar jointing, I had no idea as to why the basalt columns formed the way they did at Giant's Causeway, and now, I have my answer through this process. As well, I have concluded through my research that the Giant's Causeway was formed millions of years ago and through the splitting of two major continents: Europe and North America. Additionally, this was during a time when the Earth's mantle had active volcanic activity. Finally, I've concluded that sites like Giant's Causeway may not use the entire rock cycle to its fullest – instead taking only a small chunk of the rock cycle based on the geologic history of the site. In this case, it was magma cooling and crystallizing into igneous rock directly without any other process involved.

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